Initial Research

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13/05/2025

1 Project Directions

1. Simulation platform for running social influence models across a given network, maybe with some sort of UI to add vertices and egdges
2. Find a new heursitic for a given dataset that is more efficient than given ones from [2]
3. Visualisation for simulation of Independant and Decreasing Cascade Mdoels and allow given heuristics to see how they affect the choice of subset

1.0.1 Firmer Ideas

Abstract: Influence Maximization Evaluation Research Platform

Design and Implement a Tool for running various sampling techniques on networks, allowing for 'slot in' testing of various heuristics and comparison against existing papers. Operating on Independant Cascade, Decreasing Cascade and Linear Threashold Models.

1. Import datasets using graphml format
2. Sample (maybe 3 diff ways) and save modified subsets of data
3. Run each of the algorithms on any valid set of data
4. Probably incorporate very small sqlite db to store the data - very little overhead for reading and writing and parallelism is not important.
5. User interface CLI and exposed API with documentation
6. Export datasets as well as seed nodes and statistics - json?

2 Efficient Influence Maximization in Social Networks

[1]

1. Independant Cascade Model - Very simplistic spread of information model across a graph network, 'active' node has some probability $p_{vw} : V \times V \to [0, 1]$ for a given edge to transfer information/influence across
2. Social Network $G = (V, E)$, $S \subset V$
3. Degree Discount Heuristic is that given vertex v then if there is a direct 'active' neighbour v_{nk} of v where $deg(v) = p$ then the degree discount heursitic h would say $h(v) = p - |activeNeighbours(v)|$
4. Weights can be added to vertices to give well weight to influence
5. Running time $O(k \log n + m)$ for degree discount using fib heap.. $O(kNm)$ for greedy initial
6. Basically just runs a traversal with some extra computation

3 Learning agent influence in MAS with complex social networks

[2]

1. 14 heuristics, 5 found to be statistically significant
2. Sampling is required if dataset too large to work with given current algorithms and hardware
3. Retaining Structural Properties of the main network in the sub network is important to keep results accurate, 6 Ways listed, All have some drawbacks and upsides
4. Each of the 5 found perform differently on different networks due to underlying sub structure

4 Influential Nodes in a Diffusion Model for Social Networks

[3]

1. decreasing cascade model - 'active' nodes remain active, nodes are contagious and given some k contagion attempts on all other nodes, probability of turning a node is dependant on a function p which takes into account how many other nodes have tried to change it's state
2. Choose $S \subset V$ such that once $\forall v \in V$ v is not contagious (we have stopped computation) maximise the number of nodes that are now active

References

[1] Wei Chen, Yajun Wang, and Siyu Yang. Efficient influence maximization in social networks. In *Proceedings of the 15th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 199–208, 2009.

[2] Henry Franks, Nathan Griffiths, and Sarabjot Singh Anand. Learning agent influence in mas with complex social networks. *Autonomous Agents and Multi-Agent Systems*, 28:836–866, 2014.

[3] David Kempe, Jon Kleinberg, and Éva Tardos. Influential nodes in a diffusion model for social networks. In *international colloquium on automata, languages, and programming*, pages 1127–1138. Springer, 2005.